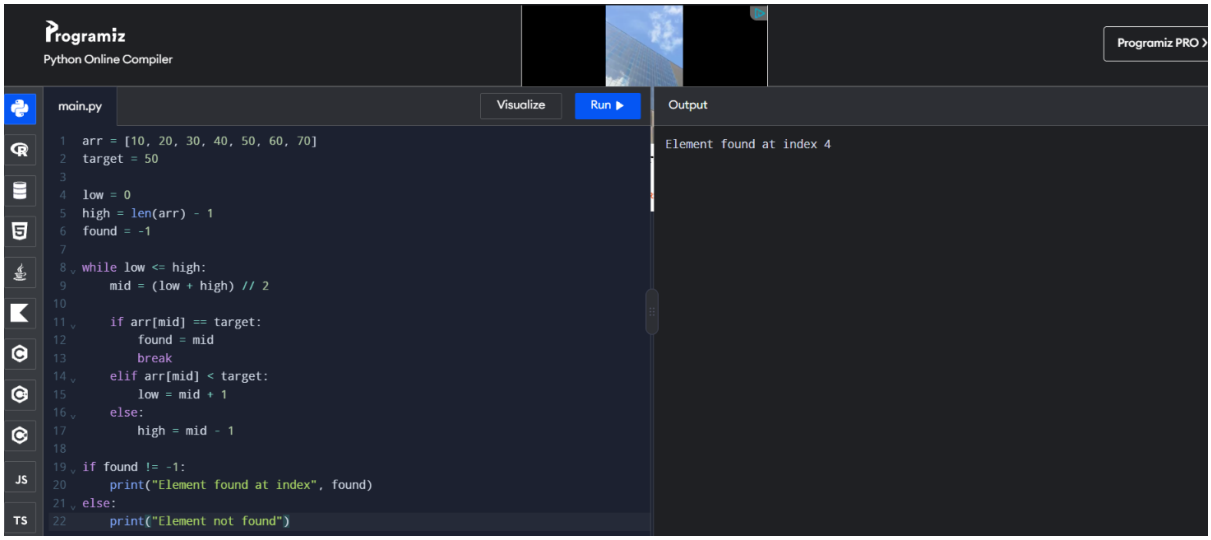
2. Champagne Tower

Aim

To determine the amount of champagne present in a particular glass of a champagne tower after pouring a given quantity into the top glass.

Procedure

1. Create a two-dimensional array to represent the glasses.
2. Pour all the champagne into the top glass.
3. If a glass contains more than 1 unit, calculate the excess.
4. Divide the excess equally between the two glasses below it.
5. Repeat the process for each row.
6. Return the champagne quantity in the required glass.



Complexity

- **Time Complexity:** $O(n^2)$
- **Space Complexity:** $O(n^2)$

Here, n is the number of rows considered.

Result

Thus, the amount of champagne present in the specified glass of the champagne tower was calculated successfully.